

Supplementary Material for "TabSurv: Tabular Foundation Model for Breast Cancer Prognosis using Gene Expression Data"

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1 Pathway Enrichment Analysis

To assess the biological coherence of the feature set used by TabSurv, we performed a pathway enrichment analysis on the 93 DRIVER_CORE genes. Of these, 90 genes were successfully mapped to Entrez gene identifiers using the org.Hs.eg.db annotation database [1] and used for subsequent analysis. The enrichment of Gene Ontology (GO) [2] was carried out in terms of the biological process, and the enrichment of the KEGG [3] pathway was carried out, with adjusted p-values using the Benjamini-Hochberg method [4] (adjusted $p < 0.05$ considered significant).

1.1 Gene Ontology Enrichment

GO Biological Process analysis identified 1,453 significantly enriched terms (adjusted $p < 0.05$). The most enriched processes were gland development (26 genes; adjusted $p = 6.02 \times 10^{-18}$), regulation of the cell cycle phase transition (26 genes; adjusted $p = 7.39 \times 10^{-18}$) and the G1/S phase transition of the cell cycle (22 genes; adjusted $p = 8.46 \times 10^{-18}$). Additional terms highly enriched included epithelial cell proliferation (25 genes; adjusted $p = 1.75 \times 10^{-16}$), radiation response (24 genes; adjusted $p = 3.41 \times 10^{-16}$), and regulation of epithelial cell proliferation (24 genes; adjusted $p = 1.06 \times 10^{-16}$). Phosphatidylinositol 3-kinase signalling was significantly enriched (14 genes; adjusted $p = 1.09 \times 10^{-12}$), consistent with the inclusion of components of the PI3K pathway in the DRIVER_CORE set. Cellular senescence (12 genes; adjusted $p = 1.71 \times 10^{-11}$) and negative regulation of the cell cycle (20 genes; adjusted $p = 6.44 \times 10^{-13}$) were also among the terms most enriched, reflecting the representation of tumour suppressor mechanisms. The top 15 enriched GO terms are shown in Figure S1a, and the complete list is available in SM-GO.

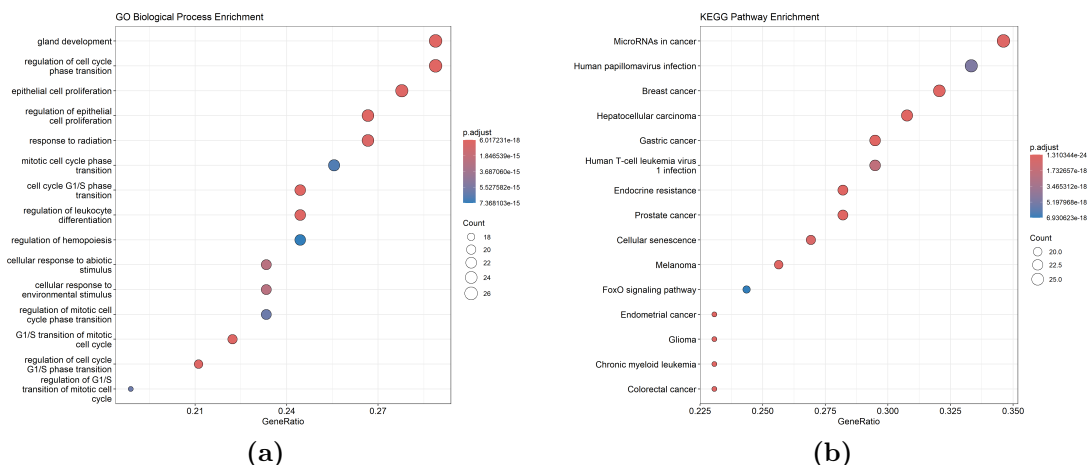


Figure S1: Pathway enrichment analysis of the DRIVER_CORE gene set. (a) Gene Ontology Biological Process enrichment. (b) KEGG pathway enrichment. Dot size indicates the number of genes; colour indicates adjusted p-value.

1.2 KEGG Pathway Enrichment

KEGG pathway analysis identified 68 significantly enriched pathways (adjusted $p < 0.05$). The most significantly enriched pathway was Breast cancer (29 genes; adjusted $p = 1.31 \times 10^{-24}$), followed by Endocrine resistance (22 genes; adjusted $p = 1.79 \times 10^{-24}$) and the PI3K-Akt signalling pathway (27 genes; adjusted $p = 3.63 \times 10^{-16}$). Additional cancer-related pathways with significant enrichment included Proteoglycans in cancer (21 genes; adjusted $p = 1.12 \times 10^{-14}$), Prostate cancer (16 genes; adjusted $p = 6.93 \times 10^{-14}$), and Hepatocellular carcinoma (17 genes; adjusted $p = 3.63 \times 10^{-13}$). Enrichment was also observed for Cellular senescence (16 genes; adjusted $p = 5.96 \times 10^{-12}$), p53 signalling pathway (11 genes; adjusted $p = 3.03 \times 10^{-10}$), and Cell cycle (13 genes; adjusted $p = 1.06 \times 10^{-9}$). Cross-cancer enrichment was evident, with significant results for Melanoma (12 genes; adjusted $p = 1.02 \times 10^{-11}$), Gastric cancer (14 genes; adjusted $p = 1.71 \times 10^{-10}$), and Colorectal cancer (12 genes; adjusted $p = 3.85 \times 10^{-10}$). The top 15 enriched KEGG pathways are shown in Figure S1b, and the complete list is available in SM-KEGG.

2 Breast Cancer Prognosis

2.1 In Distribution Experiment Results

Table S1: Benchmarking survival models under the in-distribution (InD) setting.

Dataset	TabSurv	DeepHS	DeepSurv	ENCOx	LH	MTLR	PCHazard	PMF	RSF
METABRIC	0.6265 ± 0.0079	0.5512 ± 0.0191	0.5923 ± 0.0076	0.5993 ± 0.0119	0.5872 ± 0.0218	0.5868 ± 0.0286	0.5929 ± 0.0148	0.5866 ± 0.0088	<u>0.6002</u> ± 0.0139
NKI	0.6980 ± 0.0243	0.5588 ± 0.1030	0.6572 ± 0.0601	0.6290 ± 0.0236	0.6522 ± 0.0190	0.6391 ± 0.0341	<u>0.6584</u> ± 0.0176	0.6268 ± 0.0412	0.6574 ± 0.0469

Notes: DeepHS, DeepHitSingle; ENCOx, ElasticNetCox; LH, Logistic Hazard; PMF, Probability Mass Function; MTLR, Multi-Task Logistic Regression; RSF, Random Survival Forest. Best-performing methods are shown in bold; second-best methods are underlined.

Table S2: In-distribution comparison (held-out METABRIC/NKI splits). ΔC : TabSurv minus baseline, averaged over five seeds, with 95% paired patient-level bootstrap CI; p_H : Holm-adjusted bootstrap p -value across baselines within each endpoint.

Endpoint	Baseline	ΔC (95% CI)	p_H
RFS	DeepHS	+0.0753 (+0.0606, +0.0900)	<0.0001
RFS	DeepSurv	+0.0342 (+0.0243, +0.0446)	<0.0001
RFS	ENCOx	+0.0272 (+0.0169, +0.0373)	<0.0001
RFS	LH	+0.0393 (+0.0289, +0.0500)	<0.0001
RFS	MTLR	+0.0397 (+0.0277, +0.0511)	<0.0001
RFS	PCHazard	+0.0336 (+0.0235, +0.0429)	<0.0001
RFS	PMF	+0.0399 (+0.0290, +0.0504)	<0.0001
RFS	RSF	+0.0263 (+0.0162, +0.0364)	<0.0001
DMFS	DeepHS	+0.1392 (+0.1019, +0.1753)	<0.0001
DMFS	DeepSurv	+0.0408 (+0.0114, +0.0693)	0.0090
DMFS	ENCOx	+0.0690 (+0.0421, +0.0977)	<0.0001
DMFS	LH	+0.0458 (+0.0165, +0.0773)	0.0040
DMFS	MTLR	+0.0589 (+0.0289, +0.0893)	<0.0001
DMFS	PCHazard	+0.0396 (+0.0123, +0.0695)	0.0140
DMFS	PMF	+0.0712 (+0.0413, +0.1063)	<0.0001
DMFS	RSF	+0.0406 (+0.0139, +0.0667)	0.0140

For the in-distribution scenario, each dataset is partitioned using a single random split, with 50% of the samples allocated for training and 50% reserved for testing, using five random seeds (40–44) to ensure reproducibility. We evaluate on the 2 largest cohorts, one per endpoint: METABRIC for relapse-free survival (RFS), and NKI for distant metastasis-free survival (DMFS). Table S1 and S2 demonstrates that TabSurv achieves the highest C-index on both datasets (0.6265 on METABRIC and 0.6980 on NKI), outperforming

all eight baseline survival models. The second-best method varies by dataset: RSF on METABRIC (0.6002) and PCHazard on NKI (0.6584), with TabSurv exceeding both by a margin of roughly 2–4 points in C-index.

Across both datasets, all baseline models maintain C-index values comfortably above 0.5, indicating no catastrophic failures on this benchmark; however, DeepHS shows the weakest and least stable performance, with a notably higher standard deviation on NKI (± 0.1030) compared to all other methods, suggesting sensitivity to the specific train/test split and random seed. In contrast, TabSurv maintains competitive standard deviations while achieving the top mean C-index on both datasets, potentially benefiting from the robust prior knowledge inherent in the foundation model and the proposed two-stage strategy for handling censored data.

2.2 Statistical Comparison Results

Table S3: Cohort-level ΔC (TabSurv minus baseline; mean over five seeds) with 95% paired patient-level bootstrap CIs (2,000 resamples).

Cohort	DeepHS	DeepSurv	ENCCox	LH	MTLR	PCHazard	PMF	RSF
<i>RFS</i>								
GEO	+0.091 (+0.059, +0.122)	+0.034 (+0.015, +0.054)	+0.042 (+0.023, +0.061)	+0.054 (+0.034, +0.073)	+0.048 (+0.025, +0.069)	+0.037 (+0.020, +0.054)	+0.045 (+0.023, +0.064)	+0.028 (+0.004, +0.052)
GSE19783	+0.072 (-0.019, +0.161)	+0.008 (-0.044, +0.062)	+0.039 (-0.009, +0.090)	+0.037 (-0.028, +0.102)	+0.020 (-0.042, +0.081)	+0.024 (-0.032, +0.081)	+0.040 (-0.019, +0.099)	+0.034 (-0.038, +0.111)
GSE6532	+0.092 (+0.041, +0.141)	+0.011 (-0.022, +0.045)	+0.043 (+0.017, +0.069)	+0.037 (+0.008, +0.065)	+0.036 (+0.003, +0.069)	+0.043 (+0.017, +0.067)	+0.022 (-0.009, +0.054)	+0.040 (+0.003, +0.079)
TCGA500	+0.074 (-0.022, +0.167)	+0.019 (-0.045, +0.085)	-0.012 (-0.071, +0.050)	+0.027 (-0.032, +0.086)	+0.002 (-0.069, +0.077)	+0.007 (-0.043, +0.059)	+0.031 (-0.029, +0.091)	-0.013 (-0.079, +0.051)
UK	+0.136 (+0.074, +0.197)	+0.032 (-0.006, +0.070)	+0.077 (+0.042, +0.114)	+0.063 (+0.028, +0.099)	+0.052 (+0.012, +0.091)	+0.045 (+0.007, +0.081)	+0.062 (+0.025, +0.100)	+0.046 (-0.003, +0.095)
UPP	+0.085 (+0.015, +0.154)	+0.015 (-0.028, +0.059)	+0.040 (-0.000, +0.080)	+0.024 (-0.015, +0.064)	+0.012 (-0.034, +0.061)	+0.017 (-0.016, +0.051)	-0.002 (-0.042, +0.042)	-0.012 (-0.069, +0.045)
<i>DMFS</i>								
HEL	+0.109 (+0.044, +0.175)	+0.057 (+0.005, +0.118)	+0.056 (+0.006, +0.102)	+0.048 (+0.010, +0.092)	+0.073 (+0.025, +0.125)	+0.059 (+0.017, +0.105)	+0.046 (-0.003, +0.095)	+0.026 (-0.021, +0.070)
MAINZ	+0.087 (+0.041, +0.133)	+0.067 (+0.020, +0.113)	+0.063 (+0.017, +0.108)	+0.039 (+0.004, +0.072)	+0.046 (+0.016, +0.076)	+0.040 (+0.003, +0.075)	+0.062 (+0.029, +0.094)	+0.045 (+0.012, +0.076)
TRANSBIG	+0.039 (-0.005, +0.082)	-0.016 (-0.056, +0.028)	+0.037 (-0.002, +0.075)	+0.011 (-0.022, +0.045)	+0.012 (-0.022, +0.049)	+0.015 (-0.018, +0.049)	+0.015 (+0.012, +0.086)	+0.001 (-0.030, +0.032)
UNT	+0.090 (+0.016, +0.160)	+0.031 (-0.029, +0.090)	+0.051 (+0.005, +0.101)	+0.018 (-0.032, +0.073)	+0.032 (-0.013, +0.083)	+0.020 (-0.027, +0.066)	+0.063 (+0.008, +0.119)	+0.035 (-0.011, +0.083)

Table S3 gives the paired difference in C-index, $\Delta C = C(\text{TabSurv}) - C(\text{baseline})$, averaged over the five random seeds, together with a 95% confidence interval (CI) from a paired patient-level bootstrap. In each of the 2,000 bootstrap resamples, the same patients were scored by both models across all seeds, so the intervals account for the correlation between the two models' predictions on the same patients. A positive ΔC means that TabSurv ranked patients more accurately than the baseline. These cohort-level intervals are descriptive and are not adjusted for multiplicity. Formal inference is based on the random-effects meta-analysis with Holm adjustment described in the main text.

2.3 Accuracy and Calibration of Predicted Survival

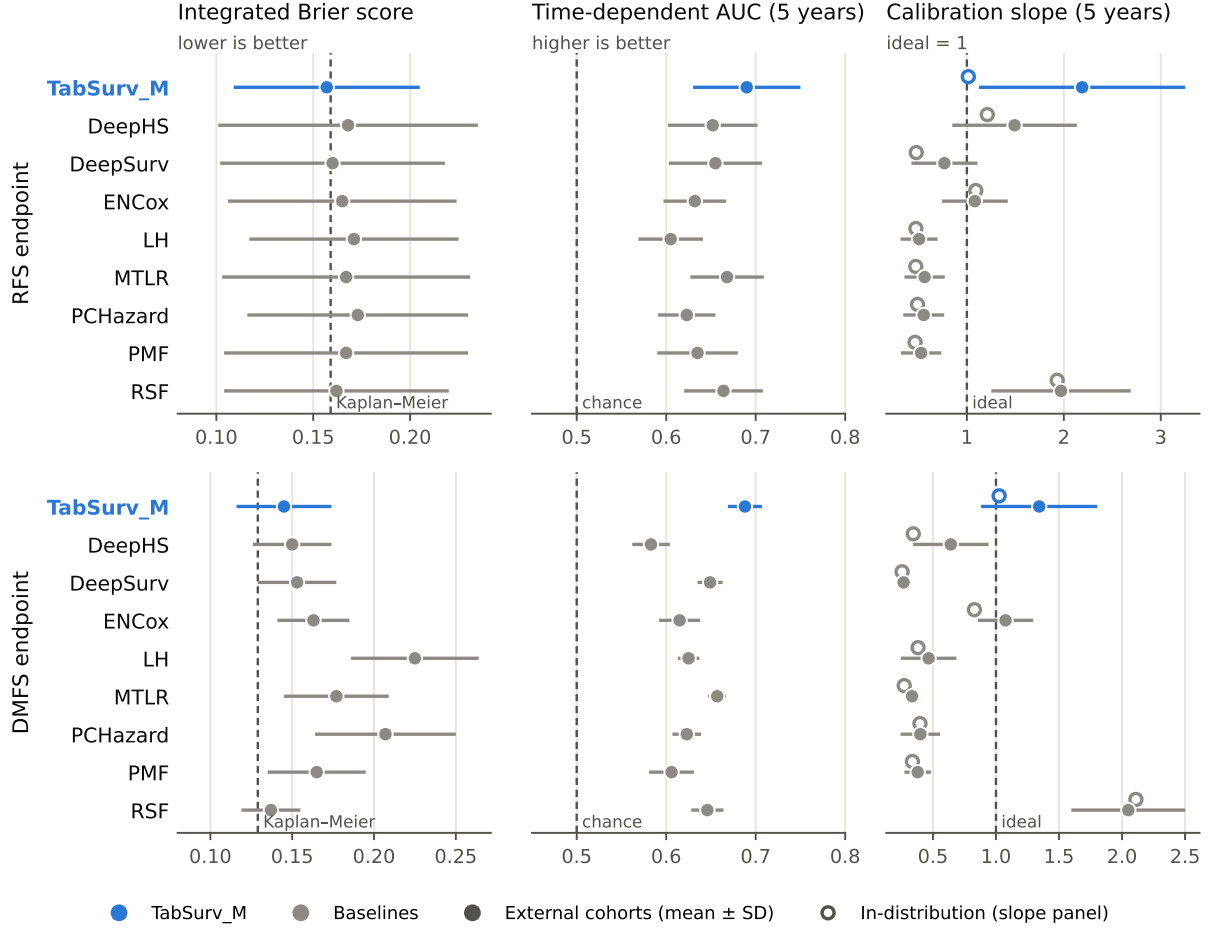


Figure S2: Accuracy and calibration of predicted survival for TabSurv and the eight baseline methods for the RFS endpoint (top; six external cohorts) and the DMFS endpoint (bottom; four external cohorts).

We additionally assessed the accuracy and calibration of each model's predicted survival function $S(t | x)$. For TabSurv, $S(t | x) = 1 - F(t | x)$, where $F(t | x)$ is obtained from the TabPFN predictive distribution (99 predictive quantiles) and averaged over the M Stage-2 models, i.e. the pooled predictive distribution across imputations. The baselines use their own survival functions; for ENCOx and DeepSurv, a Breslow baseline hazard is estimated on the training data. We report: (i) the integrated Brier score (IBS) [5], with inverse-probability-of-censoring weights estimated in each evaluation cohort, integrated from the 5th percentile of follow-up to the earlier of 10 years and the 80th percentile of follow-up, together with a covariate-free Kaplan–Meier reference; (ii) prediction-error curves (the Brier score over time); (iii) the cumulative/dynamic time-dependent AUC [6] at 3, 5 and 10 years; (iv) calibration curves at the same horizons, comparing mean predicted risk with Kaplan–Meier observed risk within quintiles of predicted risk; and (v) the calibration slope, estimated by Cox regression of the outcome on $\log\{-\log S(t_h | x)\}$ with follow-up truncated at horizon t_h [7], together with the observed-to-expected (O/E) ratio. A calibration slope of 1 indicates ideal calibration; values below 1 indicate over-dispersed predictions (risk differences between patients are exaggerated) and values above 1 indicate under-dispersed predictions. A horizon was not evaluated in a cohort whose follow-up did not extend beyond it or with fewer than 10 events before it. Predicted event times are interpreted as risk rankings, not as validated estimates of individual event times.

Figure S2 and numerical values in Table S4 compares TabSurv with the eight baselines on three measures

Table S4: Accuracy and calibration of predicted survival on the external cohorts (RFS: 6 cohorts, DMFS: 4 cohorts). IBS: integrated Brier score (lower is better), integrated up to 10 years or the 80th percentile of follow-up; the Kaplan–Meier row is a covariate-free reference. AUC(5y): cumulative/dynamic time-dependent AUC at 5 years. Slope(5y): calibration slope at 5 years (ideal 1; < 1 indicates over-dispersed, > 1 under-dispersed predictions). Values are mean $\pm$ SD across cohorts after averaging over five seeds; [k] marks metrics available in only k cohorts because follow-up did not extend beyond 5 years. Calibration curves and prediction-error curves for every cohort are provided in the Supplementary Material.

Model	RFS endpoint			DMFS endpoint		
	IBS	AUC(5y)	Slope(5y)	IBS	AUC(5y)	Slope(5y)
TabSurv_M	0.157 $\pm$ 0.048	0.690 $\pm$ 0.060	2.19 $\pm$ 1.06	0.145 $\pm$ 0.029	0.688 $\pm$ 0.019 [3]	1.34 $\pm$ 0.46 [3]
DeepHS	0.168 $\pm$ 0.067	0.652 $\pm$ 0.050	1.49 $\pm$ 0.64	0.150 $\pm$ 0.024	0.583 $\pm$ 0.021 [3]	0.64 $\pm$ 0.30 [3]
DeepSurv	0.160 $\pm$ 0.058	0.655 $\pm$ 0.052	0.77 $\pm$ 0.34	0.153 $\pm$ 0.024	0.649 $\pm$ 0.014 [3]	0.27 $\pm$ 0.02 [3]
ENCOx	0.165 $\pm$ 0.059	0.632 $\pm$ 0.035	1.08 $\pm$ 0.34	0.163 $\pm$ 0.022	0.615 $\pm$ 0.023 [3]	1.08 $\pm$ 0.22 [3]
LH	0.171 $\pm$ 0.054	0.605 $\pm$ 0.036	0.51 $\pm$ 0.19	0.225 $\pm$ 0.039	0.625 $\pm$ 0.012 [3]	0.47 $\pm$ 0.22 [3]
MTLR	0.167 $\pm$ 0.064	0.668 $\pm$ 0.041	0.56 $\pm$ 0.21	0.177 $\pm$ 0.032	0.657 $\pm$ 0.010 [3]	0.33 $\pm$ 0.03 [3]
PCHazard	0.173 $\pm$ 0.057	0.623 $\pm$ 0.032	0.55 $\pm$ 0.21	0.207 $\pm$ 0.043	0.623 $\pm$ 0.016 [3]	0.40 $\pm$ 0.16 [3]
PMF	0.167 $\pm$ 0.063	0.635 $\pm$ 0.045	0.53 $\pm$ 0.21	0.165 $\pm$ 0.030	0.606 $\pm$ 0.025 [3]	0.38 $\pm$ 0.10 [3]
RSF	0.162 $\pm$ 0.058	0.664 $\pm$ 0.044	1.97 $\pm$ 0.72	0.137 $\pm$ 0.018	0.646 $\pm$ 0.018 [3]	2.05 $\pm$ 0.45 [3]
Kaplan–Meier (reference)	0.159 $\pm$ 0.051	0.500	–	0.129 $\pm$ 0.013	0.500	–

of accuracy and calibration. The left column shows the integrated Brier score (lower is better), integrated up to 10 years or the 80th percentile of follow-up, whichever is earlier; the dashed line marks the covariate-free Kaplan–Meier reference. The middle column shows the cumulative/dynamic time-dependent AUC at 5 years (higher is better); the dashed line marks chance (0.5). The right column shows the calibration slope at 5 years; the dashed line marks the ideal value of 1, and values above or below 1 indicate under- or over-dispersed predictions, respectively. Filled markers and whiskers show the mean $\pm$ SD across external cohorts after averaging over five seeds; open markers in the right column show the in-distribution slope on the held-out METABRIC and NKI splits. The 5-year DMFS metrics are based on three cohorts because follow-up in HEL is only five years.

Numerical values for all horizons and for the in-distribution setting are given in Supplementary Table S5. Calibration and prediction-error curves for every cohort are available in the `calibration` folder of the RFS and DMFS results in the TabSurv GitHub repository (<https://github.com/vntuyen/TabSurv/tree/main/output>).

Time-dependent discrimination. TabSurv achieved the highest time-dependent AUC on both endpoints, at 3 years (RFS 0.714 vs. 0.491–0.676 for the baselines; DMFS 0.712 vs. 0.578–0.679) and at 5 years (RFS 0.690 vs. 0.605–0.668; DMFS 0.688 vs. 0.583–0.657; Figure S2, middle column), consistent with the C-index results.

Overall accuracy. On the RFS cohorts, TabSurv had the lowest IBS (0.157; baselines 0.160–0.173) and was the only model below the Kaplan–Meier reference (0.159), although by a small margin. On the DMFS cohorts, TabSurv had the second-lowest IBS (0.145, after RSF 0.137), but no model improved on the Kaplan–Meier reference (0.129; Figure S2, left column). This reflects systematic over-estimation of absolute risk in the external DMFS cohorts, whose event rates (21–31%) are lower than in the NKI training cohort (34%); the 5-year observed-to-expected ratio of TabSurv was 0.67.

Calibration. In distribution, TabSurv was well calibrated, with 5-year calibration slopes of 1.02 (RFS) and 1.03 (DMFS), the closest to the ideal value of all models (Figure S2, open markers), and the lowest IBS, below the Kaplan–Meier reference (RFS 0.159 vs. 0.173; DMFS 0.173 vs. 0.188). On the external cohorts, its calibration slope exceeded 1 (RFS 2.19 ± 1.06 ; DMFS 1.34 ± 0.46 ; filled markers): the ranking of patients transferred across cohorts, but the spread of predicted absolute risks was too narrow. The baselines also departed from ideal calibration: the neural discrete-time models were over-dispersed (slopes 0.27–0.77), RSF was under-dispersed (1.97 and 2.05), and ENCOx was closest to 1 (1.08 on both endpoints). Hence, under distribution shift, TabSurv’s predicted survival should be used for risk stratification, and absolute risk estimates should be recalibrated in the target population (e.g. by re-estimating the calibration intercept and slope) before being interpreted as individual survival probabilities.

Table S5: Accuracy and calibration of predicted survival, all horizons. IBS: integrated Brier score (lower is better), with the covariate-free Kaplan–Meier (KM) reference in each block heading. AUC(t): cumulative/dynamic time-dependent AUC at t years. Slope and O/E: calibration slope (ideal 1) and observed-to-expected ratio (ideal 1) at 5 years. In distribution: mean $\pm$ SD over five seeds on the held-out METABRIC/NKI splits. External: mean $\pm$ SD across cohorts after averaging over five seeds; [k] marks metrics available in only k cohorts because follow-up did not extend beyond the horizon.

Model	IBS	AUC(3y)	AUC(5y)	AUC(10y)	Slope(5y)	O/E(5y)
<i>RFS, in distribution (KM IBS 0.173 $\pm$ 0.004)</i>						
TabSurv	0.159 $\pm$ 0.003	0.718 $\pm$ 0.022	0.701 $\pm$ 0.009	0.668 $\pm$ 0.011	1.015 $\pm$ 0.081	1.060 $\pm$ 0.077
DeepHS	0.174 $\pm$ 0.005	0.546 $\pm$ 0.061	0.650 $\pm$ 0.014	0.582 $\pm$ 0.024	1.211 $\pm$ 0.336	1.430 $\pm$ 0.134
DeepSurv	0.176 $\pm$ 0.003	0.609 $\pm$ 0.023	0.630 $\pm$ 0.020	0.612 $\pm$ 0.018	0.478 $\pm$ 0.059	0.978 $\pm$ 0.084
ENCOx	0.172 $\pm$ 0.011	0.641 $\pm$ 0.029	0.648 $\pm$ 0.027	0.616 $\pm$ 0.016	1.092 $\pm$ 0.664	0.934 $\pm$ 0.075
LH	0.186 $\pm$ 0.004	0.542 $\pm$ 0.016	0.629 $\pm$ 0.014	0.617 $\pm$ 0.034	0.474 $\pm$ 0.047	0.938 $\pm$ 0.137
MTLR	0.177 $\pm$ 0.006	0.592 $\pm$ 0.063	0.663 $\pm$ 0.019	0.625 $\pm$ 0.030	0.473 $\pm$ 0.068	1.350 $\pm$ 0.216
PCHazard	0.190 $\pm$ 0.006	0.637 $\pm$ 0.019	0.633 $\pm$ 0.011	0.616 $\pm$ 0.020	0.491 $\pm$ 0.084	1.076 $\pm$ 0.322
PMF	0.179 $\pm$ 0.004	0.515 $\pm$ 0.075	0.642 $\pm$ 0.023	0.613 $\pm$ 0.023	0.466 $\pm$ 0.091	1.186 $\pm$ 0.176
RSF	0.166 $\pm$ 0.003	0.697 $\pm$ 0.031	0.670 $\pm$ 0.012	0.622 $\pm$ 0.026	1.932 $\pm$ 0.395	0.965 $\pm$ 0.070
<i>RFS, external cohorts (KM IBS 0.159 $\pm$ 0.051)</i>						
TabSurv	0.157 $\pm$ 0.048	0.714 $\pm$ 0.065	0.690 $\pm$ 0.060	0.620 $\pm$ 0.071 [5]	2.189 $\pm$ 1.064	0.921 $\pm$ 0.356
DeepHS	0.168 $\pm$ 0.067	0.578 $\pm$ 0.030	0.652 $\pm$ 0.050	0.565 $\pm$ 0.057 [5]	1.492 $\pm$ 0.643	1.637 $\pm$ 0.672
DeepSurv	0.160 $\pm$ 0.058	0.658 $\pm$ 0.041	0.655 $\pm$ 0.052	0.597 $\pm$ 0.070 [5]	0.768 $\pm$ 0.340	1.070 $\pm$ 0.446
ENCOx	0.165 $\pm$ 0.059	0.654 $\pm$ 0.033	0.632 $\pm$ 0.035	0.580 $\pm$ 0.047 [5]	1.082 $\pm$ 0.340	1.095 $\pm$ 0.457
LH	0.171 $\pm$ 0.054	0.518 $\pm$ 0.019	0.605 $\pm$ 0.036	0.591 $\pm$ 0.072 [5]	0.507 $\pm$ 0.191	1.117 $\pm$ 0.478
MTLR	0.167 $\pm$ 0.064	0.620 $\pm$ 0.032	0.668 $\pm$ 0.041	0.620 $\pm$ 0.065 [5]	0.564 $\pm$ 0.208	1.566 $\pm$ 0.634
PCHazard	0.173 $\pm$ 0.057	0.639 $\pm$ 0.042	0.623 $\pm$ 0.032	0.608 $\pm$ 0.046 [5]	0.555 $\pm$ 0.211	1.228 $\pm$ 0.517
PMF	0.167 $\pm$ 0.063	0.491 $\pm$ 0.042	0.635 $\pm$ 0.045	0.608 $\pm$ 0.075 [5]	0.528 $\pm$ 0.208	1.409 $\pm$ 0.579
RSF	0.162 $\pm$ 0.058	0.676 $\pm$ 0.037	0.664 $\pm$ 0.044	0.593 $\pm$ 0.067 [5]	1.971 $\pm$ 0.719	1.101 $\pm$ 0.453
<i>DMFS, in distribution (KM IBS 0.188 $\pm$ 0.005)</i>						
TabSurv	0.173 $\pm$ 0.007	0.757 $\pm$ 0.034	0.741 $\pm$ 0.033	0.725 $\pm$ 0.046	1.025 $\pm$ 0.238	1.051 $\pm$ 0.114
DeepHS	0.205 $\pm$ 0.031	0.589 $\pm$ 0.157	0.566 $\pm$ 0.123	0.569 $\pm$ 0.122	0.344 $\pm$ 0.318	1.286 $\pm$ 0.506
DeepSurv	0.218 $\pm$ 0.019	0.714 $\pm$ 0.050	0.669 $\pm$ 0.084	0.660 $\pm$ 0.095	0.255 $\pm$ 0.130	2.433 $\pm$ 1.493
ENCOx	0.206 $\pm$ 0.027	0.679 $\pm$ 0.039	0.645 $\pm$ 0.026	0.639 $\pm$ 0.057	0.829 $\pm$ 0.708	1.223 $\pm$ 0.727
LH	0.251 $\pm$ 0.090	0.650 $\pm$ 0.047	0.646 $\pm$ 0.039	0.669 $\pm$ 0.058	0.382 $\pm$ 0.209	1.424 $\pm$ 0.759
MTLR	0.226 $\pm$ 0.017	0.680 $\pm$ 0.017	0.667 $\pm$ 0.049	0.649 $\pm$ 0.084	0.272 $\pm$ 0.047	1.093 $\pm$ 0.257
PCHazard	0.232 $\pm$ 0.041	0.666 $\pm$ 0.032	0.666 $\pm$ 0.016	0.672 $\pm$ 0.055	0.397 $\pm$ 0.111	1.164 $\pm$ 0.506
PMF	0.210 $\pm$ 0.011	0.664 $\pm$ 0.094	0.627 $\pm$ 0.069	0.621 $\pm$ 0.069	0.337 $\pm$ 0.216	1.037 $\pm$ 0.229
RSF	0.182 $\pm$ 0.009	0.725 $\pm$ 0.041	0.679 $\pm$ 0.074	0.655 $\pm$ 0.074	2.110 $\pm$ 0.745	1.146 $\pm$ 0.426
<i>DMFS, external cohorts (KM IBS 0.129 $\pm$ 0.013)</i>						
TabSurv	0.145 $\pm$ 0.029	0.712 $\pm$ 0.085	0.688 $\pm$ 0.019 [3]	0.618 $\pm$ 0.040 [3]	1.343 $\pm$ 0.461 [3]	0.674 $\pm$ 0.139 [3]
DeepHS	0.150 $\pm$ 0.024	0.578 $\pm$ 0.032	0.583 $\pm$ 0.021 [3]	0.548 $\pm$ 0.015 [3]	0.642 $\pm$ 0.299 [3]	0.730 $\pm$ 0.080 [3]
DeepSurv	0.153 $\pm$ 0.024	0.655 $\pm$ 0.041	0.649 $\pm$ 0.014 [3]	0.598 $\pm$ 0.016 [3]	0.267 $\pm$ 0.020 [3]	1.806 $\pm$ 0.097 [3]
ENCOx	0.163 $\pm$ 0.022	0.636 $\pm$ 0.054	0.615 $\pm$ 0.023 [3]	0.569 $\pm$ 0.015 [3]	1.076 $\pm$ 0.219 [3]	0.743 $\pm$ 0.085 [3]
LH	0.225 $\pm$ 0.039	0.624 $\pm$ 0.037	0.625 $\pm$ 0.012 [3]	0.589 $\pm$ 0.021 [3]	0.466 $\pm$ 0.221 [3]	0.986 $\pm$ 0.100 [3]
MTLR	0.177 $\pm$ 0.032	0.661 $\pm$ 0.050	0.657 $\pm$ 0.010 [3]	0.599 $\pm$ 0.009 [3]	0.334 $\pm$ 0.029 [3]	0.687 $\pm$ 0.064 [3]
PCHazard	0.207 $\pm$ 0.043	0.600 $\pm$ 0.031	0.623 $\pm$ 0.016 [3]	0.593 $\pm$ 0.015 [3]	0.400 $\pm$ 0.157 [3]	0.783 $\pm$ 0.073 [3]
PMF	0.165 $\pm$ 0.030	0.611 $\pm$ 0.043	0.606 $\pm$ 0.025 [3]	0.563 $\pm$ 0.026 [3]	0.379 $\pm$ 0.105 [3]	0.632 $\pm$ 0.056 [3]
RSF	0.137 $\pm$ 0.018	0.679 $\pm$ 0.059	0.646 $\pm$ 0.018 [3]	0.588 $\pm$ 0.019 [3]	2.050 $\pm$ 0.452 [3]	0.644 $\pm$ 0.075 [3]

3 Treatment Recommendation Evaluation

3.1 Comparison of TabSurv Recommendation performance across 3 Gene-panels

The three established panels were also used exactly as published:

1. **Nik panel**, curated from the whole-genome sequencing study of 560 breast tumours by Nik-Zainal et al. [8]. This panel comprises protein-coding genes with strong multi-study evidence of driver activity and established roles in breast tumorigenesis.
2. **Mills panel**, based on clinically actionable and biologically relevant genes reported by Mills et al. [9], focusing on genes associated with targeted therapy response and breast cancer progression.
3. **Pereira panel**, derived from the comprehensive integrative genomic analysis of breast cancer by Pereira et al. [10], which identified recurrently mutated driver genes across a large breast cancer cohort.

Table S6: Comparison of Delta mean survival time (years) between the *Followed* and *Not Followed* groups across the three gene panels under the TabSurv recommendation framework.

Gene Set	Δ mean survival (mean $\pm$ std)
Nik panel	1.261 $\pm$ 1.992
Mills panel	0.469 $\pm$ 0.576
Pereira panel	0.693 $\pm$ 0.246

Δ Mean survival denotes the difference between *Followed* Recommendation and *Not Followed* Recommendation groups.

Table S6 summarises the difference in mean survival time between patients who followed versus did not follow the TabSurv treatment recommendation across the three investigated gene sets. The *Nik panel* set demonstrates the largest survival separation, with a mean difference of 1.261 years between the Followed and Not Followed groups. The *Pereira panel* set shows a more modest, separation of 0.693 ± 0.246 years, while the *Mills panel* set exhibits the smallest mean difference (0.469 ± 0.576 years). These results indicate that the *Nik panel* DRIVER_CORE set provides the strongest discriminatory signal for treatment recommendation within the TabSurv framework, albeit with wider variability across cohorts than the other two gene sets.

3.2 List of the 72-Gene Panel

Table S7: List of the 72 genes included in the analysis.

<i>TP53</i>	<i>PIK3CA</i>	<i>MYC</i>	<i>CCND1</i>	<i>PTEN</i>	<i>ERBB2</i>	<i>FGFR1</i>	<i>GATA3</i>
<i>RB1</i>	<i>MAP2K4</i>	<i>ZNF217</i>	<i>CDH1</i>	<i>CDKN2A</i>	<i>AKT1</i>	<i>FBXW7</i>	<i>ARID1A</i>
<i>CCND3</i>	<i>CBFB</i>	<i>MDM2</i>	<i>CCNE1</i>	<i>CDKN2B</i>	<i>NCOR1</i>	<i>SF3B1</i>	<i>SPEN</i>
<i>TBX3</i>	<i>BRCA2</i>	<i>NF1</i>	<i>PIK3R1</i>	<i>EGFR</i>	<i>ESR1</i>	<i>FOXA1</i>	<i>MED23</i>
<i>CDK6</i>	<i>NOTCH2</i>	<i>AKT2</i>	<i>BRCA1</i>	<i>CTCF</i>	<i>KDM6A</i>	<i>SETD2</i>	<i>CREBBP</i>
<i>DNMT3A</i>	<i>USP9X</i>	<i>XBP1</i>	<i>PDGFRA</i>	<i>ATR</i>	<i>ERBB3</i>	<i>FGFR2</i>	<i>RHOA</i>
<i>SMAD4</i>	<i>ATM</i>	<i>AXIN1</i>	<i>BCOR</i>	<i>CDKN1B</i>	<i>CUX1</i>	<i>GNAS</i>	<i>MLH1</i>
<i>NOTCH1</i>	<i>SMARCA4</i>	<i>STAG2</i>	<i>ZFP36L1</i>	<i>APC</i>	<i>CASP8</i>	<i>CNOT3</i>	<i>MEN1</i>
<i>MSH2</i>	<i>NRAS</i>	<i>STK11</i>	<i>ASXL1</i>	<i>CIC</i>	<i>ERCC4</i>	<i>HRAS</i>	<i>NFE2</i>

Table S7 lists the 72 of the 93 Nik driver genes that are available in the METABRIC expression data after gene matching.

3.3 Kaplan-Meier Curve of TabSurv Recommendation with Baseline Methods

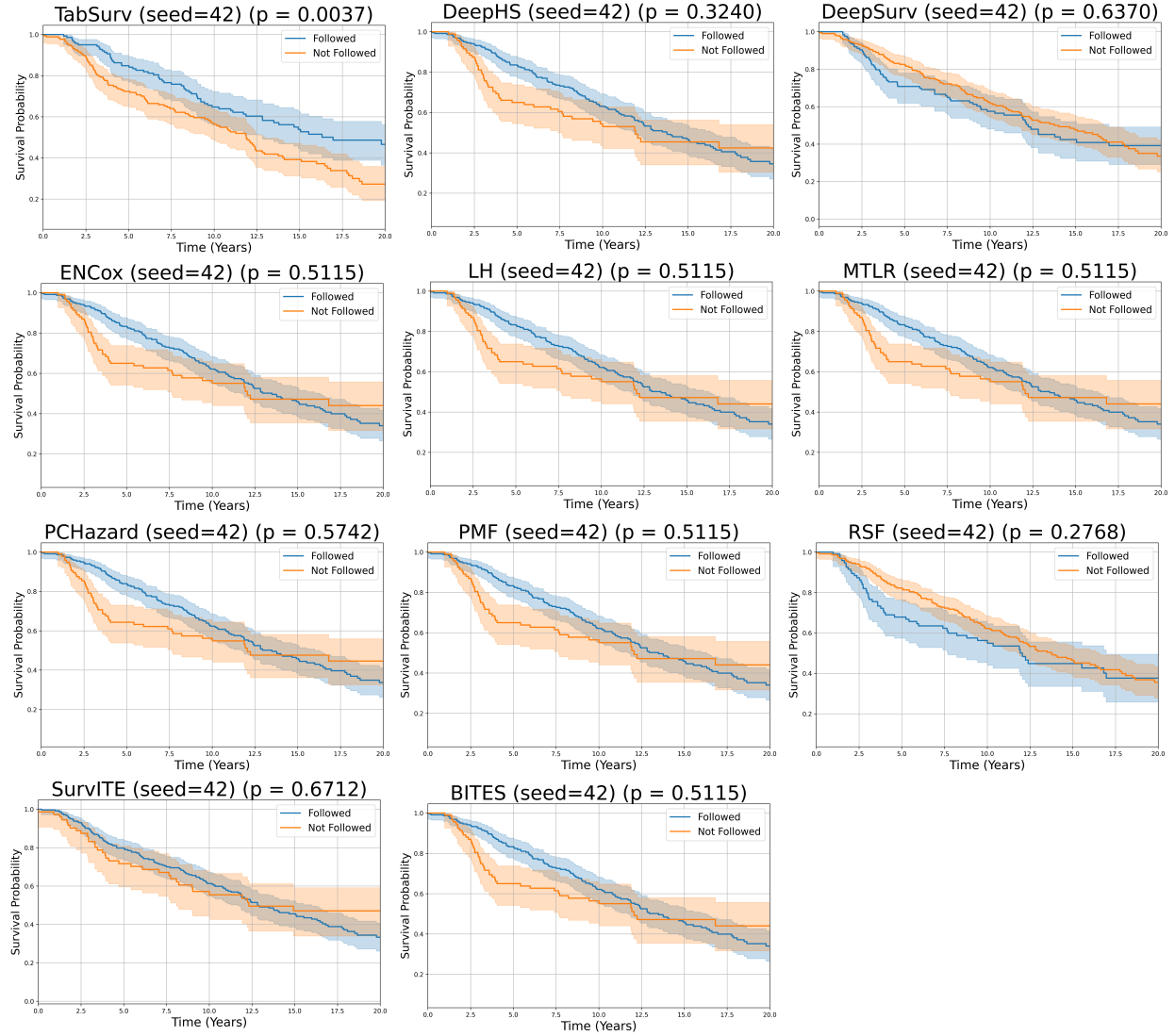


Figure S3: Kaplan-Meier survival curves for personalised treatment recommendation using selected cancer driver genes. For TabSurv and each baseline model, patients are stratified into *Followed* and *Not Followed* groups based on whether the treatment actually received matches the model-recommended therapy. Shaded areas represent 95% confidence intervals. Group differences are assessed using the log-rank test, with corresponding p -values reported.

Figure S3 presents Kaplan-Meier survival curves for all baseline methods when using the curated set of biologically important genes, compared to the TabSurv method. The curves show that, for TabSurv, the Followed group consistently exhibits a higher survival probability than the Not Followed group, whereas the baseline methods do not demonstrate a clear long-term survival benefit up to the maximum follow-up time.

References

- [1] M. Carlson, org.Hs.eg.db: Genome wide annotation for Human, r package version 3.20.0 (2024).
URL <https://bioconductor.org/packages/org.Hs.eg.db>
- [2] M. Ashburner, C. A. Ball, J. A. Blake, D. Botstein, H. Butler, J. M. Cherry, A. P. Davis, K. Dolinski, S. S. Dwight, J. T. Eppig, et al., Gene ontology: tool for the unification of biology, *Nature Genetics* 25 (1) (2000) 25–29.
- [3] M. Kanehisa, S. Goto, Kegg: Kyoto encyclopedia of genes and genomes, *Nucleic Acids Research* 28 (1) (2000) 27–30.
- [4] Y. Benjamini, Y. Hochberg, Controlling the false discovery rate: a practical and powerful approach to multiple testing, *Journal of the Royal Statistical Society: Series B (Methodological)* 57 (1) (1995) 289–300.
- [5] E. Graf, C. Schmoor, W. Sauerbrei, M. Schumacher, Assessment and comparison of prognostic classification schemes for survival data, *Statistics in medicine* 18 (17-18) (1999) 2529–2545.
- [6] H. Uno, T. Cai, L. Tian, L.-J. Wei, Evaluating prediction rules for t-year survivors with censored regression models, *Journal of the American Statistical Association* 102 (478) (2007) 527–537.
- [7] P. C. Austin, F. E. Harrell Jr, D. van Klaveren, Graphical calibration curves and the integrated calibration index (ici) for survival models, *Statistics in Medicine* 39 (21) (2020) 2714–2742.
- [8] S. Nik-Zainal, H. Davies, J. Staaf, M. Ramakrishna, D. Glodzik, X. Zou, I. Martincorena, L. B. Alexandrov, S. Martin, D. C. Wedge, et al., Landscape of somatic mutations in 560 breast cancer whole-genome sequences, *Nature* 534 (7605) (2016) 47–54.
- [9] C. Mills, A. Sud, A. Everall, D. Chubb, S. E. Lawrence, B. Kinnersley, A. J. Cornish, R. Benthams, R. S. Houlston, Genetic landscape of interval and screen detected breast cancer, *NPJ Precision Oncology* 8 (1) (2024) 122.
- [10] B. Pereira, S.-F. Chin, O. M. Rueda, H.-K. M. Volland, E. Provenzano, H. A. Bardwell, M. Pugh, L. Jones, R. Russell, S.-J. Sammut, et al., The somatic mutation profiles of 2,433 breast cancers refine their genomic and transcriptomic landscapes, *Nature communications* 7 (1) (2016) 11479.